

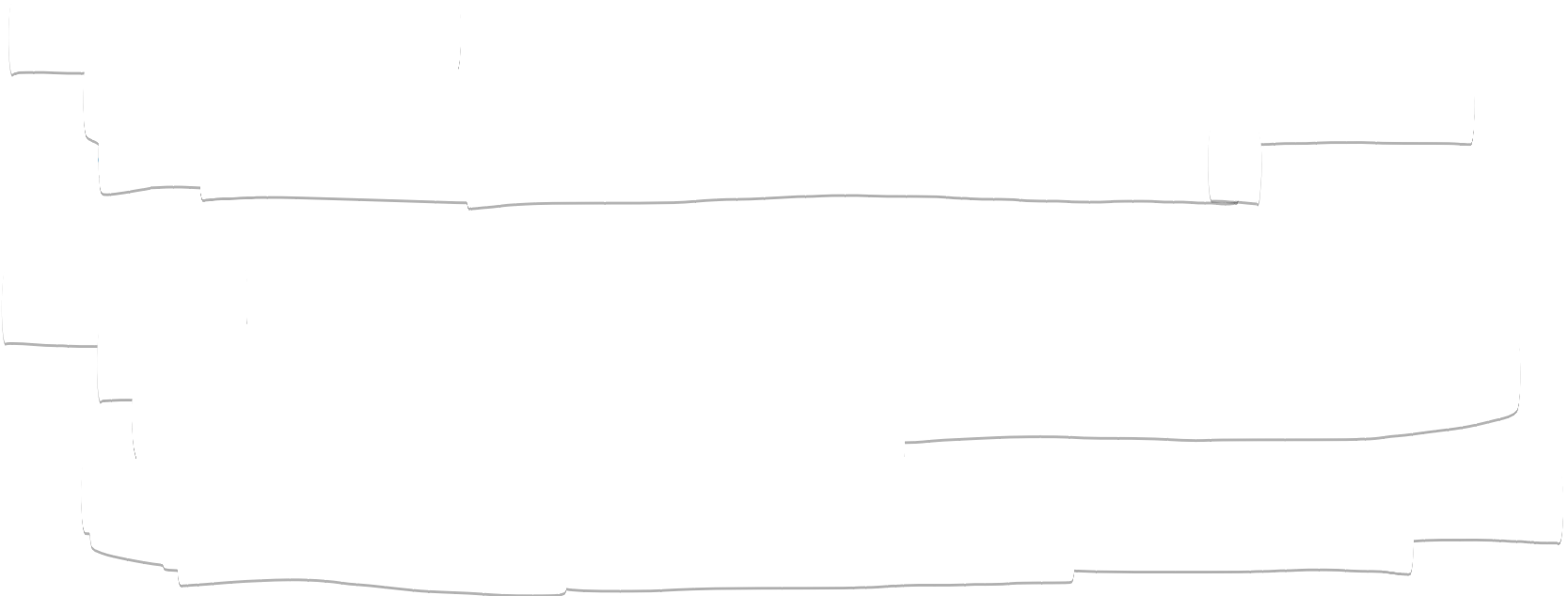


Consequences of Noise

- Major issue: *what* is noise is close to impossible to characterise without domain knowledge

Heuristics & Fuzzy Miners

- More practical approaches to process discovery
 - *Heuristic Miner*: exploits occurrence frequencies to estimate flow probabilities
 - *Fuzzy Miner*: introduces measures for significance and correlation to create abstract views of the process model



Construction of a Net System

- General net structure is directly given by the dependency graph
- Transformations needed for splits and joins, i.e., tasks with multiple incoming or outgoing edges
 - Rely on the relations between succeeding (split) or preceding tasks (join) to determine type of split/join
 - Rely on the frequencies for the split (join) and its succeeding (preceding) tasks



VALUE OF DEPENDENCY RELATIONS L RELATIONS

$ \Rightarrow_L $	a	b	c	d	e
a	$\frac{0}{0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$	$\frac{11-0}{11+0+1} = 0.92$	$\frac{13-0}{13+0+1} = 0.93$	$\frac{5-0}{5+0+1} = 0.83$
b	$\frac{0-11}{0+11+1} = -0.92$	$\frac{0}{0+1} = 0$	$\frac{10-0}{10+0+1} = 0$	$\frac{0-0}{0+0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$
c	$\frac{0-11}{0+11+1} = -0.92$	$\frac{10-0}{10+0+1} = 0$	$\frac{0}{0+1} = 0$	$\frac{0-0}{0+0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$
d	$\frac{0-13}{0+13+1} = -0.93$	$\frac{0-0}{0+0+1} = 0$	$\frac{0-0}{0+0+1} = 0$	$\frac{4}{4+1} = 0.80$	$\frac{13-0}{13+0+1} = 0.93$
e	$\frac{0-5}{0+5+1} = -0.83$	$\frac{0-11}{0+11+1} = -0.92$	$\frac{0-11}{0+11+1} = -0.92$	$\frac{0-13}{0+13+1} = -0.93$	$\frac{0}{0+1} = 0$

$|a \Rightarrow_L b|$ is the value of the dependency relation between a and b :

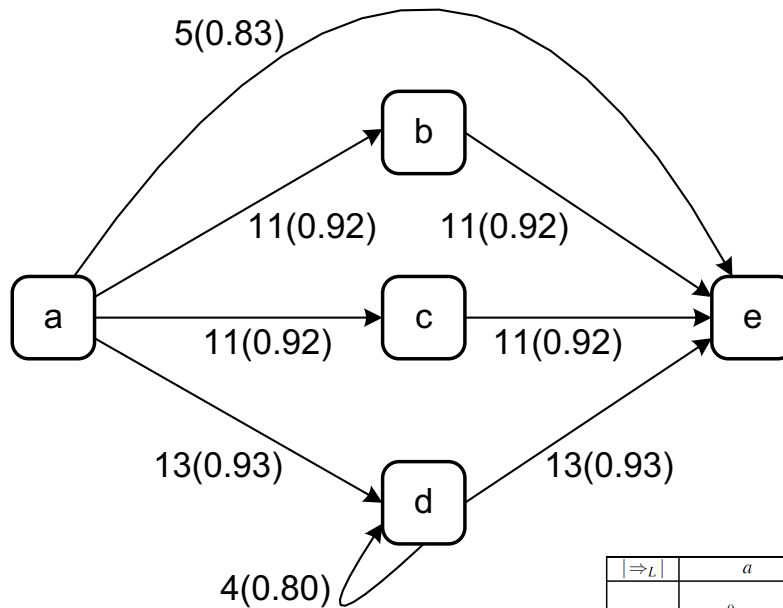
count of the direct successors

$$|a \Rightarrow_L b| = \begin{cases} \frac{|a >_L b| - |b >_L a|}{|a >_L b| + |b >_L a| + 1} & \text{if } a \neq b \\ \frac{|a >_L a|}{|a >_L a| + 1} & \text{if } a = b \end{cases}$$

$ >_L $	a	b	c	d	e
a	0	11	11	13	5
b	0	0	10	0	11
c	0	10	0	0	11
d	0	0	0	4	13
e	0	0	0	0	0

Lower threshold (2 direct succ. and a dep of ≥ 0.7)

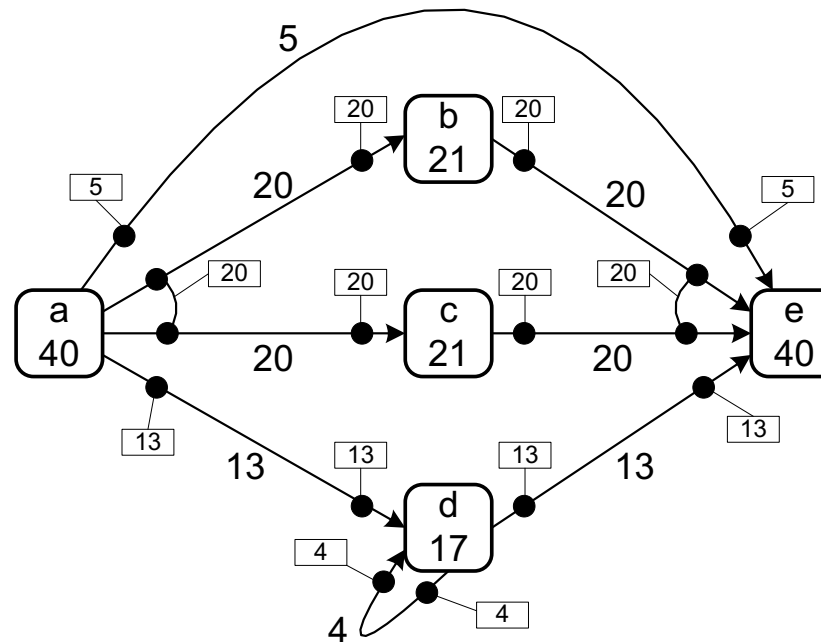
We apply a threshold to visualize only meaningful relations



$ \geq_L $	a	b	c	d	e
a	0	11	11	13	5
b	0	0	10	0	11
c	0	10	0	0	11
d	0	0	0	4	13
e	0	0	0	0	0

$ \Rightarrow_L $	a	b	c	d	e
a	$\frac{0}{0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$	$\frac{11-0}{11+0+1} = 0.92$	$\frac{13-0}{13+0+1} = 0.93$	$\frac{5-0}{5+0+1} = 0.83$
b	$\frac{0-11}{0+11+1} = -0.92$	$\frac{0}{0+1} = 0$	$\frac{10-10}{10+10+1} = 0$	$\frac{0-0}{0+0+1} = 0$	$\frac{11-0}{11+0+1} = 0.92$
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Learning splits and joins



$$L = [\langle a, e \rangle^5, \langle a, b, c, e \rangle^{10}, \langle a, c, b, e \rangle^{10}, \langle a, b, e \rangle^1, \langle a, c, e \rangle^1, \\ \langle a, d, e \rangle^{10}, \langle a, d, d, e \rangle^2, \langle a, d, d, d, e \rangle^1]$$

- Split and join rules are only considered locally (therefore most of the discovered model are not sound and require repair actions).

- Creation of views on process model that are driven by significance and correlation



- Means to cope with processing complexity
 - Aggregation: cluster similar elements
 - Abstraction (aka Projection): remove low-level information
 - Emphasis: highlight significant information
 - Customisation: adapt to context of model use

Significance and Correlation

- Metrics to control the creation of views
 - Significance of individual tasks or binary ordering relations (direct successor, causality)
 - Correlation of pairs of tasks

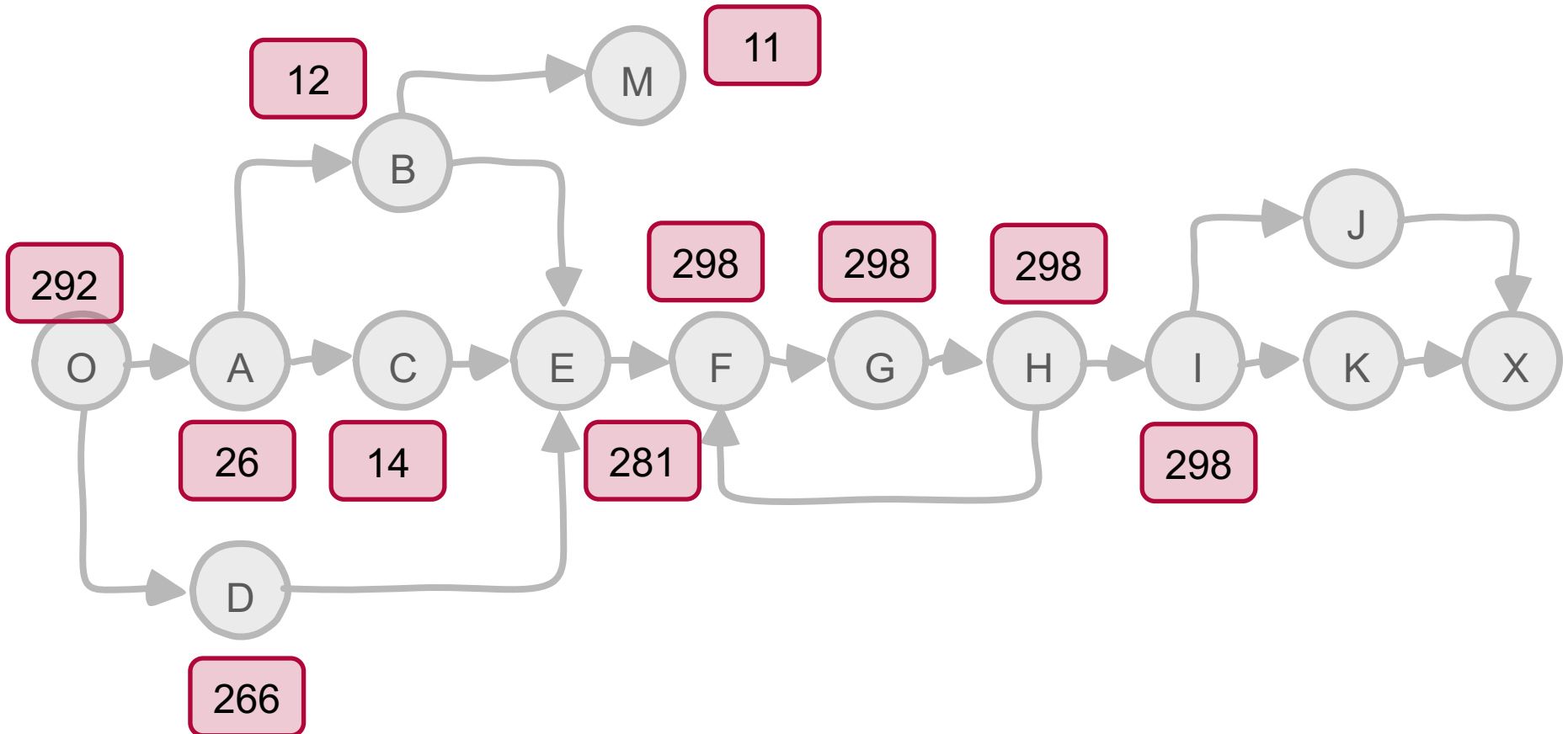
- Idea for simplification of process model:
 - Highly significant behaviour is *preserved*
 - Less significant, but highly correlated behaviour is *aggregated* into clusters
 - Less significant and less correlated behaviour is *abstracted*

Significance Metrics

- Frequency significance: Frequency of task occurrence
 - But: Can be misleading
- Routing significance: Count of distinct predecessors and successors along with their significance
 - Splits and joins typically important to understand process logic
- Relation significance: Frequency of direct successor relation for pairs of tasks
 - In addition, consider difference between relation frequency and frequency significance of source and target nodes

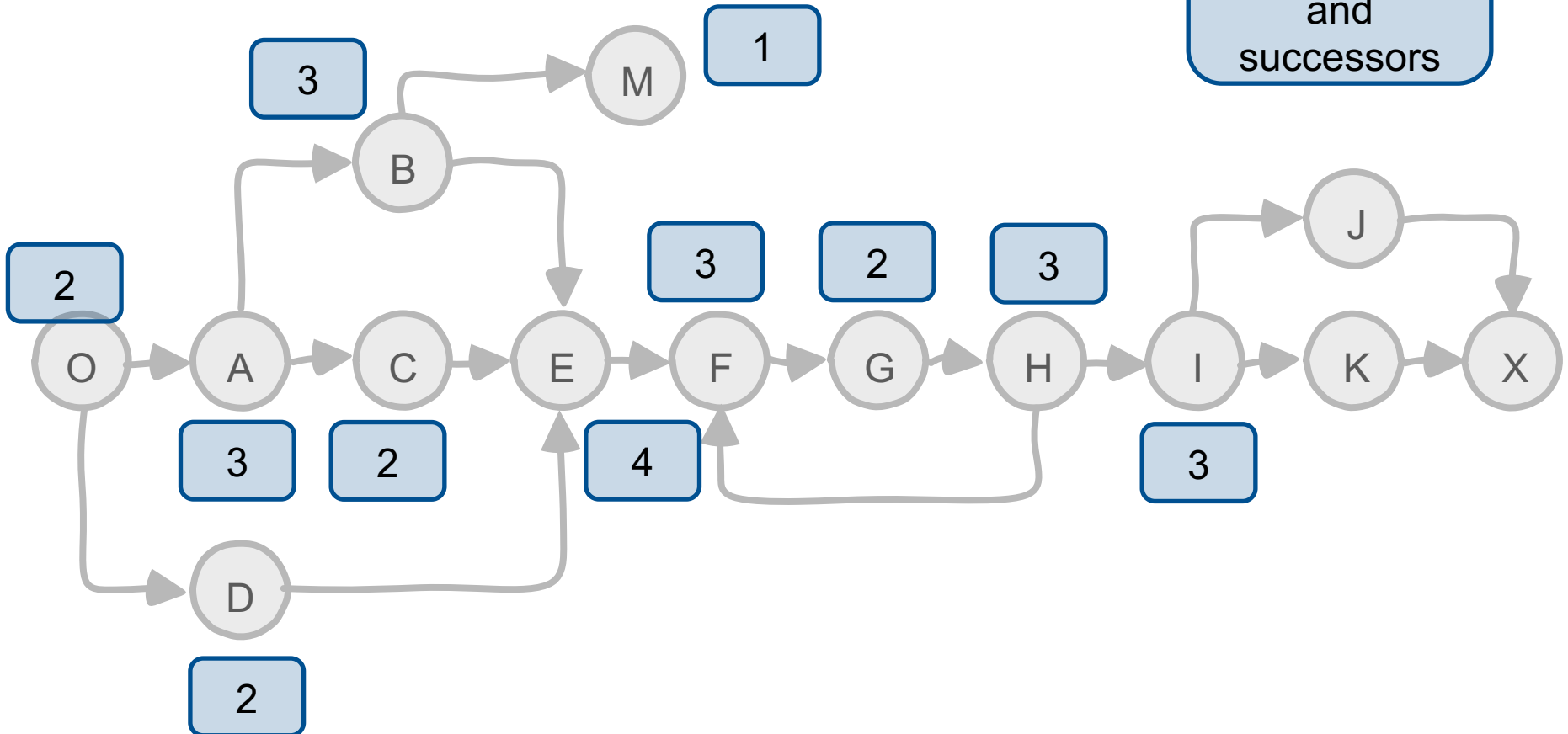
Examples

Task Frequencies



Examples

Count of
predecessors
and
successors



Correlation Metrics

- Answer the question: abstract or aggregate?
- Proximity correlation: avg distance between execution of tasks in cases



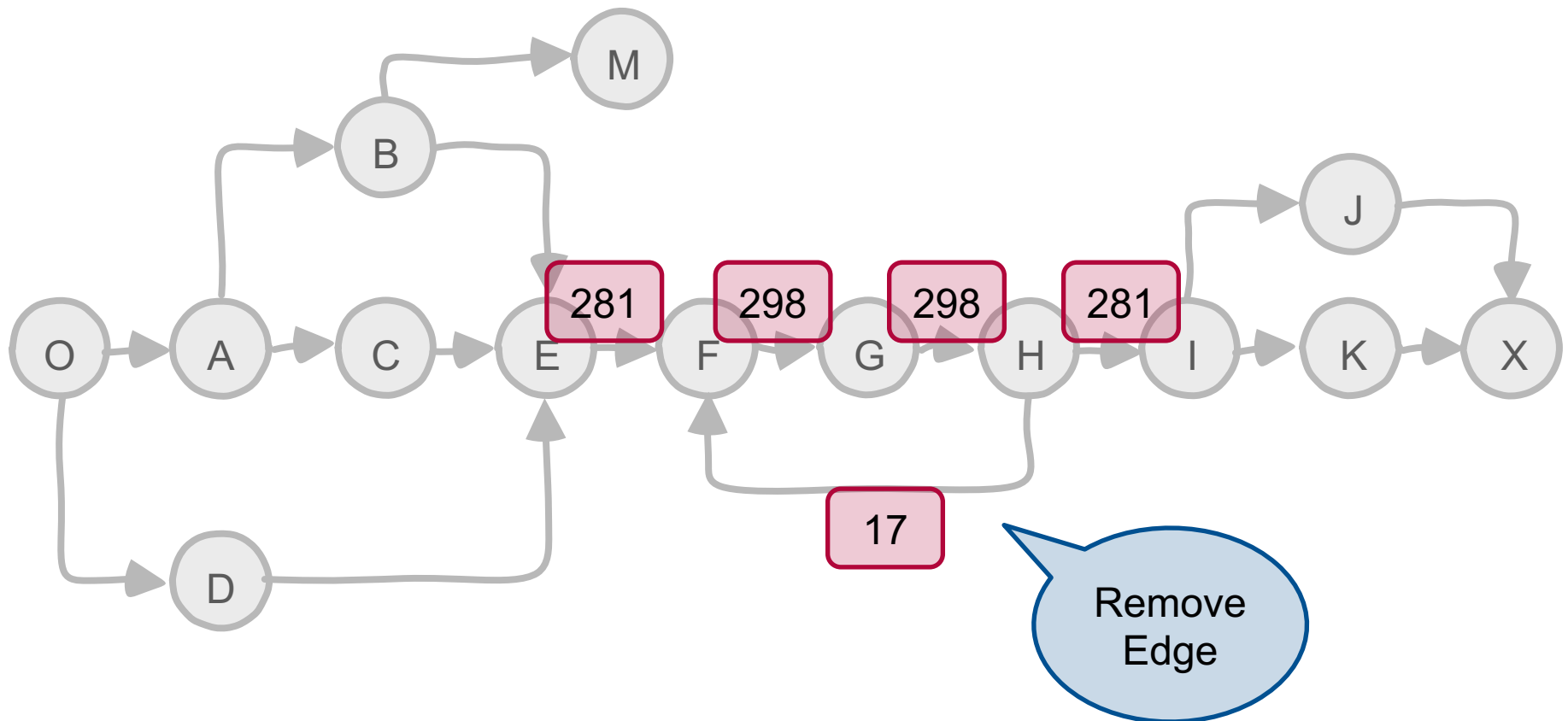
Fuzzy Miner Technique

- Start with basic dependency graph induced by causality relation of α – algorithm
- Based on significance and correlation metrics, transform graph by:
 - Edge filtering
 - Aggregation and abstraction

- **Baseline solution: Remove insignificant edges**
 - But: Tendency to create unconnected clusters of frequent behaviour
- **Thus: Compute edge utility as weighted sum of its significance and the source-target correlation**
 - For each task, maintain incoming and outgoing edge with highest utility value
 - Apart from that, filter based on threshold

Example

Edge
Frequencies



Aggregation and Abstraction

- Idea: Tasks with significance below thresholds are *victims* and are aggregated or abstracted

First phase:

- For each victim, find most highly correlated neighbour
- If neighbour is cluster, add victim
 - Cluster “inherits” incoming and outgoing edges of victim
- Otherwise, create singleton cluster with victim

Second phase:

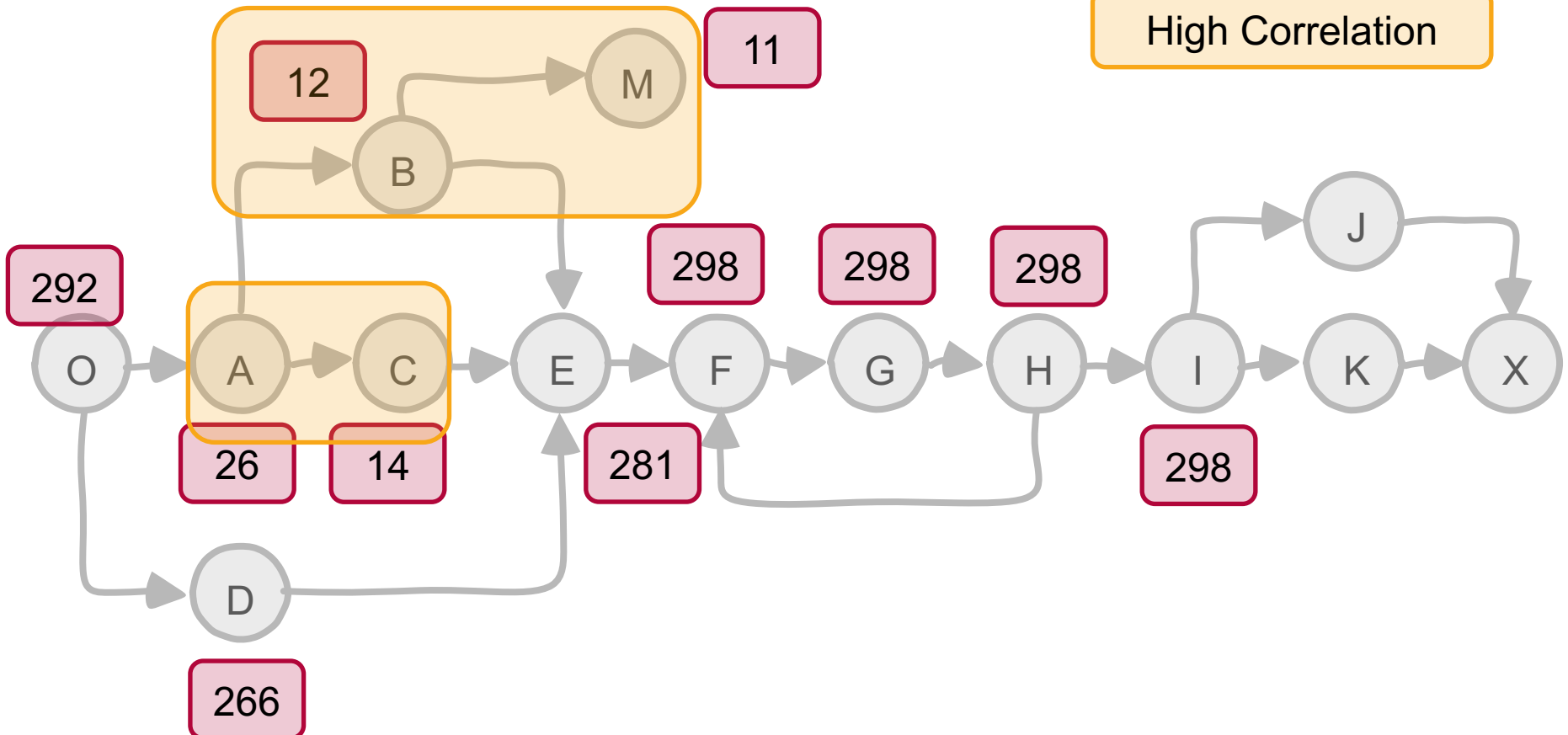
- For each cluster, check whether all predecessor and successors are also clusters
- If so, merge with most highly correlated predecessor or successor cluster, respectively

Third phase: remove isolated clusters and singleton clusters (edges are preserved transitively)

Example

Task Frequencies

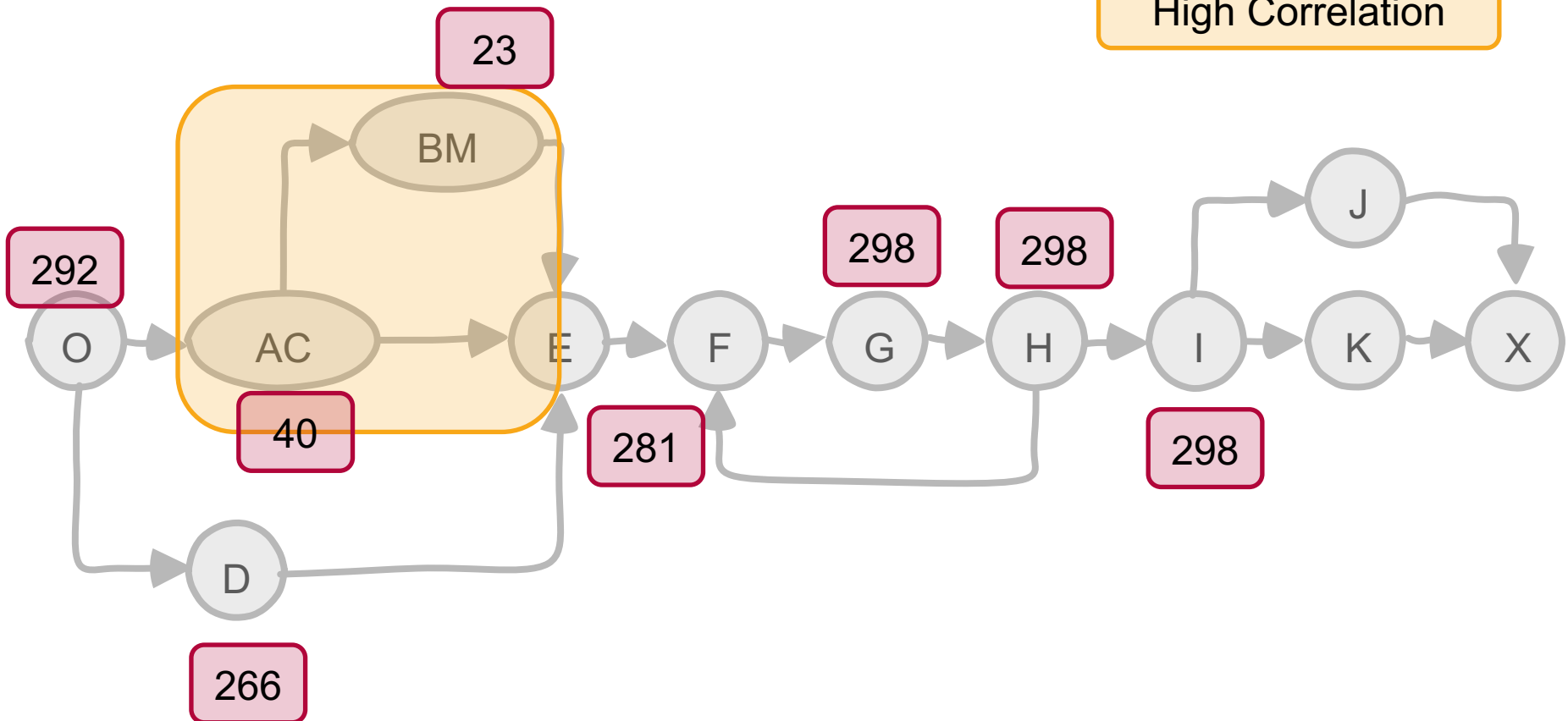
High Correlation



Example

Task Frequencies

High Correlation



Example

